

Federico Palatnik Moroz

Backend Software Engineer · API Integration · Automation Systems · Distributed Systems

- Buenos Aires, Argentina

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- federicomoroz.github.io

► Open to Remote

PROFESSIONAL SUMMARY

Backend Software Engineer specialized in automation systems, API integrations, AI-powered platforms, distributed services, and scalable backend architectures. Experienced building RESTful APIs, asynchronous workflows, webhook-based integrations, queue systems, RAG pipelines with vector search, and production-grade AI agent systems.

I integrate AI with deterministic code — guardrails, validation, and caching layers ensure LLM calls are optimal, cost-efficient, and predictable. Strong focus on clean architecture, SOLID principles, system reliability, test coverage (3,000+ tests across portfolio), and production-ready solutions.

CORE COMPETENCIES

Backend Development · API Integration · RESTful APIs · Distributed Systems · Automation Systems · Event-Driven Architecture · Queue Systems · AI Integration · RAG · Vector Search · LLM Orchestration · System Design · Software Architecture · Webhooks · Async Processing · Database Design · CI/CD · Docker · Redis · PostgreSQL · SQL · FastAPI · Flask · Python · Node.js · Microservices · Scalable Systems · Testing

TECHNICAL SKILLS

- Languages:** Python, Java, JavaScript, TypeScript, C#, SQL, PHP
- Frameworks & Libraries:** FastAPI, Flask, Node.js, React, SQLAlchemy, Alembic, Pydantic v2, dependency-injector, httpx, Spring Boot, Spring Data JPA, Spring Kafka, Hibernate, Maven, Django, Laravel
- Databases & Infrastructure:** Apache Kafka, Redis, PostgreSQL, Flyway, SQLite, Qdrant (Vector DB), Docker, Docker Compose, Gunicorn, APScheduler, Render
- AI & Integration:** Claude API (Anthropic), Voyage AI, Langfuse, RAG, Vector Search, Guardrails, LLM-as-Judge, Prompt Caching, Batch API, SSE Streaming, n8n, Airtable, BigQuery, REST APIs, Webhooks, SendGrid, Telegram Bot API
- Testing & Quality:** pytest, pytest-asyncio, pytest-cov, Playwright, mypy, Ruff, pre-commit, GitHub Actions, JUnit 5, ArchUnit, Testcontainers, architecture tests (AST and ArchUnit)
- Architecture & Practices:** OOP, SOLID, Clean Architecture, Hexagonal Architecture (Ports & Adapters), MVC, Repository Pattern, Composition Root, Event-Driven, Pipeline, Circuit Breaker, Transactional Outbox, Idempotent Consumer, Dependency Injection, Plugin Systems, CI/CD
- Development Tools:** Git, GitHub, Postman, VS Code

PROFESSIONAL EXPERIENCE

Software Developer

Freelance

2022 – Present

- Built and deployed 7+ production-grade backend services on Render, covering fintech AI agents, API gateways, distributed task queues, notification engines, and monitoring systems.
- Designed event-driven architectures using composable pipelines, pub/sub buses, circuit breakers, and repository patterns — with 3,000+ combined tests across the portfolio.
- Developed a fintech RAG agent with semantic search (Qdrant), deterministic guardrails, n8n orchestration, and LLM-as-Judge feedback loops.
- Delivered custom backend systems, automation solutions, and API integrations for international clients through Fiverr, Upwork, and direct referrals.

Systems Architecture & Automation Engineer

Del Mundo a Tus Manos

Nov 2024 – Mar 2026

- Designed and maintained a production-grade operational platform including database architecture, automation workflows, operator interfaces, and third-party API integrations.
- Reduced operational processing time from approximately 10 minutes to 5 seconds across ~2,100 monthly transactions.
- Built and maintained a production Airtable ecosystem with 11 relational tables, ~115,000 records, and 48 active automations.
- Integrated external platforms including Amazon, Mercado Libre, Tienda Nube, ARCA, messaging services, and Laravel/PHP APIs using REST APIs and webhooks.
- Developed internal operational tools including QR label generation systems and AI-assisted WhatsApp routing solutions.
- Collaborated directly with operational teams to translate business requirements into scalable technical solutions.

Game Developer

Freelance

2021 – 2024

- Developed games, gameplay modifications, and custom mechanics on commission using Unity Engine and Unreal Engine.
- Built reusable systems and modular frameworks including battle systems, AI controllers, and gameplay tools.
- Gradually transitioned toward developer tooling, focusing on backend systems and automation platforms.

Programming Instructor

Freelance

2022 – 2024

- Taught C#, JavaScript, Python, C++, Unity, and Unreal Engine to students with different technical backgrounds.
- Conducted technical assessments and created personalized study plans adapted to each student's goals.
- Mentored students through backend programming, OOP, debugging, game development, and applied mathematics.
- Provided code reviews, technical guidance, and practical project-based support.

PROJECTS

- CIRI Chargeback Agent

github.com/federicomoroz/ciri-api-aux-sourcecode · ciri-chargeback-agent.onrender.com/panel · federicomoroz.github.io/en/projects/agente-contracargos/

Fintech AI agent that automates chargeback resolution using RAG with semantic search (Qdrant + Voyage AI), deterministic guardrails ("code decides, LLM explains"), and n8n as explicit orchestrator (54 nodes). LLM-as-Judge dual-threshold feedback loop (9.1/10 score). Semantic cache for ~20% LLM cost reduction. Policy CRUD with auto-reindexing (zero downtime). Interactive testing panel with SSE streaming. Langfuse observability.

FastAPI · Claude API · Qdrant · Voyage AI · n8n · Langfuse · SQLite · 1232 Tests
- AI Training Plan Generator

federicomoroz.github.io/en/projects/mega-training-system/

Multi-discipline web platform, in daily production use, that generates professional training plans in real time using the Claude API. SSE streaming, circuit breaker on AI calls, dependency injection container, idempotency layer, plugin architecture (new disciplines without touching core), Google Drive sync, drag-and-drop class editor, and full member CRUD. PDF/CSV/HTML export.

Flask 3.0 · Claude API · SSE · SQLite · Docker · dependency-injector · Pydantic v2 · 1704 Tests
- Task Queue

github.com/federicomoroz/task-queue · task-queue-tpdz.onrender.com · federicomoroz.github.io/en/projects/task-queue/

Distributed task queue system with a FastAPI server, Redis broker (LPUSH/BRPOP), SQLite persistence, and horizontally scalable worker pool. Multi-container Docker architecture — workers scale with --scale worker=N without code changes. Supports multiple task types and retry logic.

FastAPI · Redis · SQLAlchemy 2.0 · APScheduler · Docker · 37 Tests
- Order Outbox Service

github.com/federicomoroz/order-outbox-service · federicomoroz.github.io/en/projects/order-outbox-service/

Two independently deployable microservices on a Maven reactor. Transactional Outbox on the producer side: the event is written in the same transaction as the order and a publisher dispatches it to Kafka, so an order cannot exist without its event. On the consumer side, an idempotent consumer discards redeliveries by event key. The hexagonal architecture is not asserted — ArchUnit verifies it, and the test fails if a layer imports one it has no business importing. Flyway migrations and integration tests with Testcontainers.

Java 21 · Spring Boot · Apache Kafka · PostgreSQL · Flyway · ArchUnit · Testcontainers · JUnit 5 · 72 unit tests
- Shipping Quote

github.com/federicomoroz/shipping-quote · shipping-quote.onrender.com · federicomoroz.github.io/en/projects/shipping-quote/

Shipping rate quoter with an end-to-end hexagonal architecture: every request emits its own ordered trace —entry, adapter, port, use case, domain and back out— and that trace is returned in the response. Three simulated carrier adapters on a single class configured by composition; one fails on purpose and the use case answers with two quotes out of three. Money in Decimal with an explicit ROUND_HALF_UP.

FastAPI · SQLAlchemy · Alembic · httpx · Pydantic · pytest-asyncio · 31 Tests
- Rate Guardian

github.com/federicomoroz/rate-guardian

API gateway with sliding-window rate limiting (Redis sorted sets), Redis-backed circuit breaker (CLOSED/OPEN/HALF_OPEN), transparent HTTP proxy, and event-driven logging via pub/sub bus. The gateway never imports the logger directly — fully decoupled via EventManager.

FastAPI · Redis · Docker · Event-Driven Architecture · 202 Tests
- Notify Router

github.com/federicomoroz/notify-router

Multi-channel notification routing engine with configurable rules and event dispatching to Email (SendGrid), Telegram, Slack, and generic webhooks. Pipeline + Strategy + EventManager architecture. Single-page admin UI for managing channels, rules, and events without CLI access.

FastAPI · SQLite · httpx · APScheduler · SendGrid · Telegram Bot API
- Dollar Tracker

github.com/federicomoroz/dollar-tracker

Automated exchange rate monitoring service that polls live Argentine dollar rates, stores historical data, fires email alerts when prices cross user-defined thresholds, and delivers configurable periodic reports (hourly, daily, or weekly) — fully unattended.

FastAPI · PostgreSQL · SQLAlchemy · APScheduler · SendGrid · Docker
- Webhook Logger

github.com/federicomoroz/webhook-logger · webhook-logger-9paz.onrender.com

Backend service for receiving, persisting, and auditing webhook events from any source (GitHub, Stripe, MercadoLibre, etc.). Stores payload, filtered headers, source IP, and timestamp. REST API for full event history and a built-in outbound HTTP request tester.

FastAPI · SQLite · SQLAlchemy · httpx · MVC + Repository Pattern
- envCheck

github.com/federicomoroz/envcheck

Zero-dependency CLI tool that validates a .env file against .env.example — reports missing variables, empty values, and undeclared keys before they cause runtime errors. Supports custom file paths via CLI flags.

Python · CLI · Zero Dependencies
- RPG Turn-Based Battler System

github.com/federicomoroz/RPG-Turn-Based-Battler

Modular, data-driven turn-based battle framework built in Unity (URP, 2D). Drop-in architecture lets developers add new skills, units, and controllers without touching existing code. Implements a 4-phase FSM (Anticipation → Execution → Outcome → Resolution), a Chain of Responsibility damage pipeline, parry-window system with reactive player input, configurable projectile system, status effects, and a generic 3-type object pooling system with reflection-based auto-discovery of poolable types.

Unity · C# · URP · FSM · Template Method · Strategy · Chain of Responsibility · Object Pool

EDUCATION

Escuela Da Vinci — Programming

2021 – 2023

UADE — Marketing

2007 – 2010

LANGUAGES

- Spanish

Native
- English

Full Professional Proficiency (C2)